

Living World Model

A practical architecture for persistent characters, events and consequences

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Abstract

Most game worlds become quiet outside the player's immediate surroundings. A distant non-player character (NPC) may effectively stop living until the player returns, and a location may be rebuilt from a few saved facts. This limits stories in which people make plans, relationships change and consequences travel through the world while the player is elsewhere. Interactive films face the same problem when viewers can leave one storyline and later return to another.

The Living World Model is a proposed architecture for making such worlds persistent without simulating every person and object in full detail at every second. It stores the world's changing facts in a shared, time-aware memory. It then places limited areas of attention around the player, important NPCs and active events. We call each such area a **Simulation Field**. A field is a temporary working set: it says which people, objects, places and ongoing events must be considered together right now. It can follow a player, remain with an important NPC or connect distant characters during a phone call.

Each field advances through **Beats**: ordered updates that record what changed and why. Nearby action may use two-second Beats with detailed perception and body movement. A remote workday may use one-hour Beats and record only decisions that matter later. Characters keep distinct records of what actually happened to them, what they perceived, what they remember and what they believe. Several fields can run at different speeds, then synchronize safely when their stories meet.

The report explains the ideas first in everyday language and then as data contracts, pseudocode, city examples and a prototype plan. It is a design proposal rather than a report of measured performance or psychological realism.

Contributions

The proposal makes five design commitments. First, all characters and objects share one authoritative history. Second, a Beat separates facts that affect the world from optional descriptive detail. Third, detail has several independent settings: time-step length, physical detail, social and mental detail, and presentation detail. Fourth, fields synchronize causes and consequences, not only clock times. Finally, storytellers can suggest dramatic opportunities, while characters still act from their own knowledge, needs and intentions.

The practical objective is a small, testable living world that can later grow without changing what a Beat, a belief or an accepted fact means. The numeric examples are proposed defaults, not measured results.

1 Why a living world needs selective simulation

A player enters a cafe. Elsewhere in the city, a minister answers questions and Devon negotiates a change of shift. Only the cafe is on screen, but the other events may still matter. The minister's answer could appear on a news feed. Devon could arrive later with a new plan. If these events are invented only when the player meets them, earlier facts may conflict. Simulating every remote gesture, however, would waste time and compute.

The Living World Model spends detail where it can change an interaction. The cafe may need exact speech, posture and perception. Devon's ordinary week may need only schedules and a few decisions. Both still use the same identities, world clock and history. This makes the proposal useful as a persistence and story layer for computer games, interactive films, multiplayer worlds and visual world models.

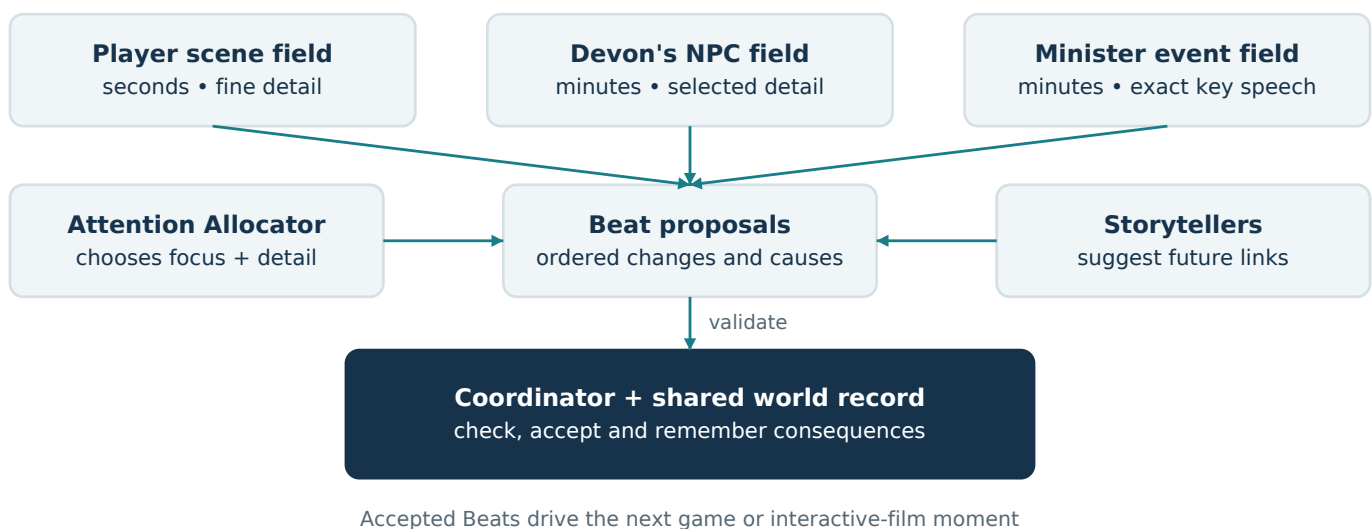


Figure 1. Several Simulation Fields advance different parts of the same world. Each produces Beats, which are checked and stored in one shared world record. The game or interactive film presents only accepted results. The Attention Allocator decides where detail is needed; storytellers suggest possible connections between plots.

1.1 What happens during one update

The Attention Allocator chooses the relevant entities and detail. A local simulator proposes their next Beat. The coordinator checks time, ownership, world rules and each character's access to information. Accepted changes enter shared history, then the game engine, dialogue system or visual world model presents the result.

A storyteller may propose that Devon receives a clue, but the coordinator checks whether it can reach him. A character model may propose a belief only from something that character perceived, remembered or inferred. A Beat is a meaningful world update, not a video frame: animation may run 60 times per second while social simulation advances every two seconds.

1.2 What this architecture adds

The proposal is a coordination and memory layer for models and game systems. Sections 2-5 define its vocabulary and character experience; Sections 6-12 explain detail, fields and synchronization; Sections 13-15 cover story direction and a prototype.

A reader's guide to the main ideas

At 14:00 the player leaves City Hall. At 14:10 the minister makes an off-screen statement. At 14:20 Devon hears a distorted report. At 14:30 the player asks Devon what happened. A convincing answer requires the system to know what was said, what reached Devon and what he understood.

The Living World Model treats this as a problem of selective simulation and persistent memory. It advances only the relevant part of the world at useful detail, but it keeps every important consequence in a shared history.

Term	Plain-language meaning	Why it exists
Entity	Anything with a persistent identity	Alice, a ring, a cafe, a faction or an idea
Agent	An entity whose perception and inner state are simulated	A player character, NPC or creature
Event-Entity	A scene or longer activity that exists through time	A conversation, journey, trial or investigation
Beat	One ordered package of changes during a time interval	Records who did, learned, felt or changed what
Simulation Field	The temporary set of entities and events updated together	Focuses compute around a player, NPC or consequential event
Attention Allocator	The scheduler that assigns detail, time steps and compute	Spends resources where mistakes would matter most
Storyteller	A planner that proposes future dramatic opportunities	Connects local events into longer plots without rewriting facts

A Simulation Field is a moving spotlight, not a bubble

Picture a film crew moving through a city. It records the player scene in detail, keeps a smaller crew with an important off-screen NPC, and receives summaries from elsewhere. The spotlight can follow relationships and communication as well as distance. A phone call can connect distant fields; a promised delivery can pull a courier into context before arrival.

Its **active set** contains state it may update. Its **context set** contains facts it may read. Its **dependency set** contains remote events that must synchronize before it advances. This is why a field is more than a radius on a map.

What “more detail” means

Detail is not one quality slider. The system can choose a shorter time step, richer physical description, deeper social reasoning, more precise beliefs, or more presentation detail. These settings form a small list of choices. Technical literature sometimes calls such a list a vector; this report names the settings directly so their meaning remains clear.

2 Related work and design boundaries

This proposal combines ideas from story memory, changing graphs, simulated agents and distributed systems. The closest work helps explain the parts; none of the cited systems is assumed to provide the complete Living World Model described here.

2.1 Narrative and temporal memory

The Narrative World Model stores changing story information in a graph and retrieves the parts relevant to a writing task [1]. A graph is a network of named things and labeled relationships; temporal means that facts can change over time. The Living World Model uses a similar kind of memory inside a simulation loop. It adds the proposed scheduler, Beat protocol, autonomous fields and synchronization rules described here.

Dynamic-graph research studies networks whose things and relationships change; the survey by Kazemi et al. provides broader context [3]. In this report, a **Temporal World Knowledge Graph** simply means the world's time-aware factual memory. It does not require a particular AI embedding method or graph database. A prototype can store the same information in ordinary database tables and follow indexed relationships.

2.2 Generative agents and sensory worlds

Generative Agents integrates recorded experiences, reflection, retrieval and planning in an interactive sandbox [4]. This report adds explicit separation between canonical participation, subjective observations and later recollection, along with contracts for partial simulation and distributed ownership. These are engineering choices for this proposal rather than claims that one cognitive theory fully describes human behavior.

Genie 3's original announcement describes interactive environment generation and identifies extended interaction and multi-agent interaction as limitations [2]. The Living World Model addresses a complementary persistence layer. Integration with a visual world model would require an interface that maps accepted symbolic changes into observable audiovisual changes and reconciles deviations. It is a proposed integration, not an available capability established here.

2.3 Causal ordering

Lamport distinguishes causal order from an arbitrary ordering of distributed events [5]. The point is intuitive: two unrelated conversations can happen in either processing order, but a message must be sent before it is received. The architecture records such cause-and-effect links and uses a stable tie-break rule only when replay needs one. A shared clock alone cannot prove that a background field has produced every earlier effect.

2.4 What the proposal does not promise

The architecture cannot guarantee consistency if generators can invent unregistered communication channels or retroactive effects. It cannot make a bounded context contain an arbitrarily complex scene. It does not equate graph completeness with psychological completeness. Its useful guarantees are conditional on a declared ontology, controlled boundary channels, verified transition rules and an honest record of what has not been simulated.

The MVP therefore uses a central coordinator and a small closed world. Distribution is an extension after the contracts work, not a prerequisite for testing the central idea.

3 One shared world, many histories and viewpoints

Every persistent thing has an identity. People, objects, locations, groups and even abstract concepts can be entities. The Temporal World Knowledge Graph connects them with labeled, time-stamped facts: Alice holds the ring from 14:00 to 14:23; Devon works for the cafe during June; the conference takes place at City Hall. Looking up one entity gathers its changes and relationships without creating a second copy of the world.

Term	Meaning in this report	Example
Entity	Persistent identity with typed state and relations	Alice, a ring, City Hall
Agent	Entity with modeled perception and internal state	Devon, a minister
Event-Entity	A temporally extended activity or occurrence	Press conference, conversation, journey
Process Entity	Optional subtype for ongoing development	Investigation, epidemic, marital conflict
Beat	One logical simulation tick and its transition record	A question is asked and a denial follows
Change	An ordered update inside a Beat	Ring possession transfers to Devon
Snapshot	Materialized state after accepted changes	Cafe state at 14:23:17
Narrative thread	Authorial representation of an arc or promise	Will the minister resign?

A scene is normally an Event-Entity; a conference can contain interviews, and an investigation can span many fields. Concepts also have history: recycling is a concept node, Alice's attitude toward it is a temporal belief, and a campaign promoting it is an Event-Entity.

3.1 An event and a field answer different questions

An Event-Entity answers, “What is happening?” A Simulation Field answers, “What must we update together right now?” One field may contain a cafe conversation and a delivery. One large event, such as a citywide protest, may span several coordinated fields. Moving an event between fields changes who computes it, not what the event is or what already happened.

3.2 Facts need time and a source

A stored fact records what it concerns, what it says, when it is true and which Beat supports it. The system also records when it saved the fact and when a character learned it. Those times can differ: a secret may be true on Monday, stored by the world on Monday and discovered by Alice on Thursday. **Provenance** is the source trail that explains where a fact or belief came from.

Use half-open validity intervals such as [14:00, 14:10). Absence means unknown or unmodeled, not false. Plans and candidate futures live outside the canonical fact scope. Historical queries specify both the world time and, where needed, the record version being reconstructed.

4 Beats: the world's change receipts

A Beat is one meaningful update to the simulated world. Think of it as a receipt listing all changes caused during an interval. A close conversation might produce a Beat every two seconds. A sleeping household might use a single Beat for several hours. The Beat stores changes; a **snapshot** is a convenient saved picture of the resulting state. Keeping them separate prevents the same change from being applied twice when detail is added later.

4.1 What every Beat must say

Each Beat covers a start and end time. Within it, **microsteps** preserve order when several things share the same clock reading: Alice speaks, Devon hears her, then Devon reacts. Independent changes receive a stable replay order, but the system does not pretend one caused the other. The contract below is a logical record design; an implementation may rename or regroup its fields.

```
Beat
  id, field_id, owner_epoch, event_ids
  interval: (t0, t1], base_snapshot_version
  participants: entity IDs and roles
  resolution: level, component overrides, temporal precision
  coverage: entity x component -> evaluation status
  changes: ordered Change records
  inbound_message_ids, outbound_messages
  causal_parent_ids, unresolved_detail_constraints
  model_and_rule_versions, seed, validation_receipt
  status: proposed | committed | rejected

Change
  id, subject_id, component_path, scope
  effective_time, microstep, causal_parent_ids
  operation, expected_prior_version, new_value
  evidence_ids, precision, supersedes_or_refines
```

4.2 Complete logging at the chosen detail

Every modeled change must be logged, whether it affects an object, place, person, group, event or idea. Unchanged values need not be copied. The record must still distinguish “checked and unchanged” from “not examined at this level of detail.” Otherwise a later system could wrongly assume that silence proves nothing happened.

Continuous quantities may be represented by a validated trajectory or transition function with parameters, rather than a sample at every millisecond. Preserve turning points and intermediate changes that affect observations or downstream decisions. An agent who becomes alarmed and then calm within a Beat must not lose that causal history merely because the endpoint matches the start.

4.3 Atomicity and projection

A commit accepts the Beat's changes, scheduled boundary messages and ownership checks together. All entity views reference these same records. If a handoff transfers a ring, both possession views derive from the same transition. Inconsistent duplicate writes are rejected. Presentation is generated from the accepted Beat; a prose scene cannot independently introduce a new canonical action.

4.4 Worked Beat at the cafe

At 14:23:15, Alice holds a ring. Devon has just arrived. Alice hands it over while saying, "Keep it until tomorrow." Devon has an active fictional sedative effect, making attention less reliable. The cafe has a quiet back room, and a delivery negotiation is also underway. The next Beat ends at 14:23:17.

Order	Entity and component	Accepted change or record
15.4 / 0	Ring / physical custody	Alice to Devon; title of ownership unchanged
15.4 / 1	Alice and Devon / relation views	Possession projections update from the ring transfer
15.5 / 0	Alice / speech act	Requests safekeeping until tomorrow; utterance recorded
15.5 / 1	Devon / perception	Hears request; source is Alice's speech, not hidden intent
15.8 / 0	Devon / intention	Accepts safekeeping; target and deadline recorded
16.0 / 0	Devon / posture	Closes hand around ring; relevant to visibility
16.2 / 0	Alice / emotion	Concern decreases after Devon's acceptance
16.5 / 0	Safekeeping / Event-Entity	Enters active phase; return obligation is open
17.0 / 0	Devon / effect state	Rule advances sedative duration to the Beat endpoint

The times are illustrative exact simulation times, not a psychological measurement. A rule-driven engine may supply them. A coarse generator may instead propose an interval; that interval must be refined before a time-sensitive observer can depend on the ordering.

What the Beat does not assert

Devon does not automatically know why Alice wants the ring hidden. Alice does not automatically know how clearly Devon heard her. His sedative effect does not justify an arbitrary false belief; it changes an explicit attention or encoding rule. The delivery group acquires no shared knowledge simply because one member was present.

The cafe's occupancy is evaluated unchanged in this Beat. Its building condition is not modeled. The related marital conflict is read but receives no new state unless the exchange actually affects it. Thus the log distinguishes participants, context dependencies and changed entities.

Why this is stronger than a scene summary

"Alice gave Devon the ring and felt reassured" is useful prose, but does not specify custody versus ownership, perception, obligation, timing or effect duration. The Beat contract retains those distinctions while allowing the prose to remain short. A summary points to this Beat; it does not replace its authoritative changes.

5 What happened, what I noticed and what I believe

For a character, “what happened to me” can mean several different things. Alice can attend a conference without hearing every sentence. She can hear a sentence without remembering it accurately. She can also believe something she never witnessed because another person told her. The architecture stores these layers separately and links each conclusion to its sources.

Layer	Question answered	Example
Participation	What happened involving Alice?	Alice attended the conference
Perception	What reached her senses and attention?	She heard a partial sentence through noise
Memory trace	What was encoded and retained?	A fragment linked to a nervous tone
Recollection	What did she reconstruct later?	She recalls a stronger denial than was spoken
Belief	What did she accept or suspect at time t?	She suspects the minister hid evidence

Presence is not perception. Perception is not comprehension. Memory is not an exact transcript. Belief need not be a memory at all: an agent may infer something never witnessed. A false belief is a real internal state whose proposition is false; it is not a correction to world history.

5.1 World uncertainty and character uncertainty

The system may not yet have decided a minor world detail, and a character may be unsure about a belief. These are different. The accepted world history—often called **canon**—can say that the ring is in the cafe while Alice confidently believes it is at home. Conversely, the storyteller may leave a past detail open even though Alice already has a definite belief about it. Later elaboration must respect what she experienced.

5.2 Subjective history remains temporal

Memory traces can weaken, be reinterpreted or acquire later associations. Store such changes as versions rather than editing the original perception. Each recollection references its sources, retrieval time, reconstructed content and confidence. A recollection may update a belief, but merely retrieving a memory does not have to rewrite it.

5.3 Scope and access

The world resolver uses hidden canon to determine sensory access: sight, hearing, smell, taste, touch, balance, proprioception and internal bodily sensations. The character model receives only its own perceptual packet, accessible memories and internal state. Other agents' private beliefs are excluded, not supplied with an instruction to ignore them. Separate contexts protect against accidental omniscience.

For groups, distinguish an official position, shared records and a distribution of member beliefs. A faction's public policy is not what every member privately thinks. A collective mind exists only if the fictional setting explicitly defines one.

5.4 How an agent acquires a belief

A character forms beliefs only from information available to that character: perceptions, memories, testimony and earlier beliefs. A language model may interpret ambiguous evidence, but it should not read the answer from hidden world state and then invent a matching reaction. The following pipeline turns this principle into a proposed behavioral policy.

```

UPDATE_AGENT(agent, accepted_world_changes, time)
  percepts := PERCEIVE(changes, senses, access, attention, effects)
  memories := RECALL(agent.memory, current cues, retrieval budget)
  evidence := percepts + memories + prior accessible testimony
  candidates := INTERPRET(evidence, prior beliefs, goals, traits)
  for each candidate:
    CHECK all claimed evidence was accessible to this agent
    IDENTIFY support, contradiction and shared source ancestry
    APPLY configured trust and belief-revision policy
    KEEP uncertainty if evidence does not resolve alternatives
  APPRAISE revised beliefs against needs, goals and commitments
  PROPOSE emotion, intention and salient thought changes
  VALIDATE provenance and append internal deltas to the Beat

```

Example of a mistaken conclusion

The minister says, "I did not authorize the payment." Alice hears only "authorize the payment" and sees him hesitate. Her perception records the missing audio. Because she already suspects misconduct, she provisionally interprets the fragment as evidence of involvement. Her belief becomes "the minister probably authorized it," with its source and uncertainty attached. The canonical speech record remains unchanged.

Later Devon plays the full recording. Alice now has contradictory evidence. She may retract the authorization belief while retaining a separate suspicion that the minister concealed someone else's involvement. The model must represent that distinction rather than flipping a single global trust score. If she rejects the recording, the refusal needs an accessible reason, such as prior evidence of edited clips.

Confidence policy

For the MVP, use ordinal confidence such as low, medium and high with explicit support and contradiction lists. Do not treat a model's self-reported 0.83 as a calibrated probability. A quantitative version may use Bayesian or other evidence updates only where its assumptions and likelihoods are defined. Multiple reports copied from one broadcast count as correlated testimony, not independent confirmations.

Motivation can bias interpretation under a declared character model; it cannot manufacture sensory access. A hallucination or magical illusion must have a modeled cause and a subjective perception record, not an unmarked alteration of the objective scene.

5.5 Internal state and selective Theory of Mind

Characters need enough inner state for their actions to remain understandable over time. The system does not need to invent a complete inner monologue on every Beat. It records only state that may affect behavior: needs, emotions, intentions, salient thoughts, beliefs, bodily condition and relevant effects.

Component	Required structure when represented
Physical state	Position or trajectory, health, movement constraints, fatigue
Social state	Roles, commitments, attributed relationships, public status
Emotion	Type, target, cause or appraisal, intensity band, duration rule
Need	Dimension, level, threshold, rate rule, available responses
Intention	Goal, chosen commitment, prerequisites, status, deadline
Thought	Salient content, trigger, active interval, causal relevance
Posture	Pose or gesture category, visibility, action constraints
Effect	Source, target, onset, expiry, modifiers, stacking policy
Belief	Holder, proposition, temporal scope, stance, evidence, confidence

An effect can represent a spell, intoxication or a fictional drug. Its modifiers may affect perception, movement, needs or memory encoding. Expiry is a scheduled transition even if narrative detail is low. This is a game-mechanics contract, not a pharmacological model. Physiological rules must not be guessed from medical-sounding descriptions.

Beliefs about beliefs

Theory of Mind here means representing what one character thinks another character knows, wants or believes. In compact notation, $B(A, p)$ means “Alice believes p .” $B(A, B(\text{Bob}, p))$ means “Alice believes that Bob believes p .” A third level can represent what Alice thinks Bob believes Charlie believes. These attributed mental states may be wrong; Bob's actual belief remains a separate record.

Create higher-order records when an active decision, observed interaction or explicit thought requires them. Alice may infer that Bob thinks Charlie knows about the ring. That record needs evidence available to Alice and may be wrong. The actual Bob-belief record stays separate. Unknown attributed belief is not disbelief. Missing depth-three state does not mean the agent considered and rejected it.

At the highest detail level, eligible higher-order beliefs are checked for relevance, not generated exhaustively. A default maximum depth of three is a resource policy, not a claim about cognitive limits. If a pivotal deception needs deeper reasoning, the allocator can grant an override or return a documented reasoning limitation.

Store fictional thoughts as authored state outputs, not as a purported transcript of the language model's private reasoning. The simulator needs their behavioral consequences and provenance, not token-by-token introspection.

6 Six levels of semantic detail

A **semantic detail level** is a checklist of what the simulator must represent when relevant. It is separate from Beat length and from what is shown on screen. A high-stakes statement five kilometers away may require exact wording but no posture. A nearby silent chase may require precise motion with almost no dialogue. The six profiles below give teams a shared vocabulary; individual components can be promoted when needed.

Level	Default semantic contract	Typical exclusions
L0 Aggregate	Population and regional state, major scheduled changes, all external effects	Individual dialogue and uninstantiated minds
L1 Outline	Named participants, activity phases, essential outcomes, obligations, consequential transfers	Incidental gestures and routine thoughts
L2 Causal	Ordered causes and consequences, evidence routes, salient perception, changed beliefs, goals, needs and effects	Verbatim routine dialogue and fine posture
L3 Social	Decision-relevant memory, emotion appraisal, intentions, testimony, relevant depth-two beliefs	Exhaustive internal narration
L4 Scene	Relevant exact speech acts, fine actions, spatial and perceptual constraints, salient bodily state	Unobserved decorative microdetail
L5 Embodied	Relevant multisensory experience, posture, gestures, active thoughts, detailed appraisal; depth-three beliefs when needed	Full physical microstate and unbounded Theory of Mind

6.1 Rules that every level must obey

Even the coarsest profile must preserve changes that affect other parts of the world: messages, object transfers, conflicting uses of a resource, access to information and timed effects. If an outline says Devon became suspicious and that suspicion changes his next action, the system must record what he suspects and why. Lower detail saves descriptive work; it does not excuse missing consequences.

Lower detail means fewer distinctions are instantiated, not permission to erase distinctions already committed. Existing high-resolution facts remain queryable after a field is demoted. An omitted component is marked unmodeled, not unchanged. Fine states required by the renderer or engine may be stored outside the language context and accessed through typed queries.

6.2 Per-component overrides

Profiles are defaults. A Beat may use L2 overall, exact speech for one statement, L4 visibility and L1 background crowd activity. The effective contract is the union of required fields and overrides. Promotion adds detail within existing commitments; demotion changes future computation only. The allocator reports which requirements could not be satisfied rather than silently claiming L5 coverage.

An L5 Beat can span ten seconds; an L1 Beat can span one second. Neither numbering nor duration determines the other. The shortest necessary causal boundary, not the prose length, determines when a Beat must stop.

7 The Attention Allocator: where should the world think harder?

The Attention Allocator is the resource manager for Simulation Fields. It answers four questions: What must be included? How much detail does each part need? How soon must the field advance? How much compute may it spend? A dramatic character may receive more attention, but narrative importance never overrides a physical fact or an already scheduled consequence.

7.1 First collect plausible candidates

Begin with the player, important NPCs, participants in active events, nearby entities and anything with a scheduled consequence. Then follow useful relationships such as possession, promises, direct perception, communication, hazards and active plot threads. A broad link such as “resident of the city” must not load every resident. The search limits which relationship types it follows, how many steps it takes and how many results each step may add.

7.2 A visible relevance score

After mandatory entities are included, optional candidates can be ranked. For candidate e and field F , each signal ranges from 0 to 1: s is spatial closeness, g is strength of an approved relationship path, c is causal importance, n is narrative importance, i is expected interaction and u is uncertainty that could change the next action. One inspectable starting formula is:

```
R(e,F) = 0.20s + 0.15g + 0.25c + 0.15n + 0.15i + 0.10u
s = exp(-travel_time(e, anchor) / spatial_scale)
g = maximum path weight over allowed typed paths
path weight = product of configured edge weights
```

This is an inspectable heuristic, not a learned optimum. Mandatory dependencies bypass the threshold entirely. Optional candidates are admitted by score subject to cost; seed thresholds can be L1 below 0.30, L2 at 0.30, L3 at 0.50, L4 at 0.70 and L5 at 0.90. L0 describes aggregate substitutes rather than a forced identity downgrade. Component requirements can raise any level.

7.3 Budget and scheduling

First reserve compute for mandatory contracts. Allocate the remainder to upgrades with the highest estimated reduction in decision or continuity error per unit cost. Maintain distinct limits for input tokens, output tokens, model calls, latency and memory. A long input can be expensive even when the desired Beat is only a short summary.

Schedule fields using interaction deadlines, unresolved causal dependencies, NPC autonomy priorities and an increasing staleness term. Give dormant but important NPCs a minimum service allocation so attention does not collapse around players. Keep token reservations for interrupts and synchronization. Hysteresis and a minimum residence time prevent a moving player from causing repeated promotion and demotion at a field edge.

7.4 How a Simulation Field chooses its reach

A Simulation Field is often drawn as a circle around a player, but the useful boundary follows interaction. Its **active set** contains state the field may change. Its **context set** contains facts it may read. Its **dependency set** contains remote activity that must be synchronized before the field advances. Reading a city summary does not give the field permission to change the whole city. Calling a minister does not require detailed simulation of every street between both phones.

A spatial starting point, plus non-spatial links

Distance remains useful for discovering people or objects that may soon interact. The field starts loading them early enough to finish before contact. A proposed **prefetch radius** is:

```
r_prefetch = r_contact + v_bound * latency_bound + margin
```

Here `r_contact` is the distance for immediate sensing or contact. `v_bound` is the fastest allowed closing speed. `latency_bound` is the measured time needed for catch-up, generation and validation. The margin covers uncertainty. Streets, doors, sightlines and audibility may matter more than straight-line distance. Teleportation and phone calls use explicit triggers instead of forcing the radius to cover the whole world.

The final field is the union of mandatory dependencies and budgeted optional candidates, including non-spatial links. Its spatial footprint may be irregular. Choose the smallest footprint that covers required physical interactions; there is no requirement for a single circle. Expand further only when benefit justifies the cost.

```
RESOLVE_FIELD(anchors, time, horizon, budget)
  candidates := SPATIAL_PREFETCH + TYPED_GRAPH_CANDIDATES
  required := PARTICIPANTS + SAFETY + DUE_BOUNDARIES + READ_DEPENDENCIES
  repeat:
    ADD prerequisites of required transitions and selected actions
    RESOLVE cross-field owners or subscribe to boundary channels
  until dependencies are closed or an explicit limit is reached
  if a required dependency is unresolved: SHORTEN, SYNCHRONIZE or WAIT
  ASSIGN minimum component contracts to required entities
  FILL remaining budget with scored candidates and detail upgrades
  EMIT field sets, levels, horizon, cost and exclusion reasons
```

An LLM can propose semantic dependencies, such as a debt behind a confrontation. A deterministic resolver checks entity IDs, expands approved relations and enforces budgets. It must not mistake a plausible dependency suggestion for verified fact.

If required state alone exceeds the context, shorten the horizon, stream exact tool queries, or partition independent work under a shared coordinator. A tightly coupled scene may genuinely exceed capacity. Slowing simulation is preferable to silently dropping the ring or a critical belief. Dependency closure is bounded by the modeled world, not magically complete in an unrestricted universe.

7.5 Worked allocation in a city

The player is running toward Cafe North at 14:23:17. The minister has an autonomous field at City Hall. Devon is in a nearby transit concourse. A city director tracks a council dispute. World context includes a curfew policy, transit disruption and the current weather, each as a versioned summary with exact constraints where needed.

For this illustrative preset, immediate contact and sensing cover 30 meters. A relative approach bound of 10 meters per second, an assumed five-second processing bound and a 20-meter margin give a 100-meter prefetch radius. A loud alarm or a pending phone call is added even if its source is farther away. If the measured processing bound rises, the radius or prefetch lead time must rise too, or movement must wait at an interaction boundary.

Context or field	Chosen detail	Preferred Beat	Why
Player and cafe participants	L4, L5 gesture override	2 seconds	Immediate visible interaction
Concourse and Devon	L3	15 seconds	Likely arrival and an active commitment
Minister at City Hall	L3, exact public statements	60 seconds	Independent consequential NPC plot
City services	L0 summaries; exact closures	5 minutes	Broad context with scheduled effects
Remote regions	L0 versioned summaries	6 hours or triggers	No near-term modeled interaction

These are maximum preferred step sizes, not permission to cross a message or hazard boundary. A sedative expiring in seven seconds splits Devon's 15-second Beat. A live broadcast may require immediate exact exports even while private conference detail remains coarse.

An example compute allocation

Suppose the player field has a 12,000-token input limit. Reserve 5,000 for mandatory current state and constraints, 2,000 for recent Beats, 1,000 for world/city summaries and 3,000 for optional historical evidence. Keep 1,000 unfilled for an interrupt. A separate 2,000-token output limit covers proposed changes and presentation. These values are test settings, not latency claims; measuring them is part of the prototype.

Devon is not promoted merely because he is nearby. His incoming route and commitment make him relevant. A passerby one meter away may remain shallow unless interaction makes posture, identity or belief important. Conversely, a minister five kilometers away becomes a high-resolution participant immediately when a call connects. Spatial distance predicts relevance but never defines it alone.

8 Building the information packet for a field

Before a model can propose a Beat, the system builds a **context packet**: the small, relevant part of world memory the model may use. It first loads required current state, active Event-Entities and world rules. It then adds recent Beats, trusted summaries and older evidence relevant to the present question. “Complete” means complete for the promised components, not every fact ever stored about every character.

8.1 One relationship step or several

A **graph hop** means following one stored relationship. From the ring, one hop may find its current holder, location or latest transfer. It does not load the holder's entire biography. “What does Alice think Bob knows about the ring?” may require several steps through belief, evidence and perception records. Retrieval rules should name the relationships they need instead of assuming every question has the same hop count.

Keyword search finds exact names and phrases. Semantic search finds passages with related meaning, commonly by comparing numerical representations called **embedding vectors**. These searches may suggest additional context, but stable entity IDs and required facts never depend on search rank. The system follows only approved relationships, limits large fan-outs and removes duplicated evidence.

8.2 A different packet for each viewpoint

The world coordinator can inspect authoritative facts. Each character model receives only that character's accessible evidence and internal state. A narrator receives accepted observable results plus the chosen narrative viewpoint. The Meta-Storyteller may see broad world summaries, but that knowledge does not automatically appear inside a character's mind.

```
ContextPacket
  field_id, world_time, snapshot_version, authority_scope
  required_entities_and_components, current_event_states
  hard_constraints, pending_boundaries, active_effect_deadlines
  recent_beat_refs, bounded_history_summaries
  perspective_packets: agent_id -> accessible_evidence + state
  optional_retrieved_evidence, omissions_and_coverage
  output_contract, token_limits, unresolved_dependencies
```

8.3 History is compressed without losing provenance

Summaries carry the interval they cover, source Beat IDs, perspective, last update and unresolved questions. Current inventory, resource reservations and hard commitments come from exact projections, never solely from prose summaries. A director's summary can say “trust is deteriorating,” while a character packet still distinguishes Alice's suspicion from Bob's actual conduct.

On demand, the system retrieves older evidence beneath the summary. Generated memories of unmodeled experience are labeled elaborations and checked against existing history. Retrieval cannot make a character remember something that has no authorized path into that character's experience or prior knowledge.

9 Keeping fields at different speeds consistent

Different Simulation Fields may use two-second, one-minute or one-day Beats. They still share one fictional clock and one accepted history. Real processing time is separate: if a model takes five seconds to answer, that does not mean five seconds passed for a character unless the simulation says so.

Time marker	Meaning
T	Target world time requested by the game or server
c(F)	Field's latest committed state frontier
d(F)	Frontier elaborated at the field's current detail level
W(G,F)	Logical time through which G has sealed its outputs to F
t_contact	World time at which an inter-field interaction occurs

A **frontier** is simply how far a field has safely progressed. A field may have coarse state through 14:23:17 but detailed narration only through 14:22. More detail does not mean more elapsed time. Exact frontiers also include a microstep, because reaching 14:23:17 is different from finishing every immediate reaction at that same time.

9.1 The late-message problem

Imagine the cafe has accepted events through 14:25 while the conference is only simulated through 14:20. During later catch-up, the conference invents a public resignation broadcast at 14:22. Cafe patrons should already have reacted, so the histories now conflict. Sorting messages afterward cannot repair reactions that were never simulated.

Before a field accepts history through time t , every modeled source of incoming effects must have delivered everything through t or certified that nothing more can arrive from that earlier period. This certificate is called a **boundary watermark**. In plain language it says, “This channel is complete through here.” The receiving field must also process everything already delivered through that point.

9.2 Advance consequences before decorative detail

A background field can decide its outward effects in a coarse pass while leaving private dialogue and harmless gestures unspecified. Later refinement may add texture, but cannot invent an earlier phone call or broadcast after other fields have moved past it. If a source cannot safely bound its effects, it must advance together with the receiving field. “Probably nothing important happens” is not enough.

This design preserves dormant-world efficiency only where the boundary contract is enforced. In an unrestricted world where any sleeping NPC may secretly affect anyone at any time, safe lazy synchronization is impossible without rollback or stronger restrictions.

9.3 Boundary channels and the next safe stopping point

A **boundary channel** is any modeled way one field can affect another: a person crossing a doorway, sound reaching a room, a phone message, a broadcast or use of a shared object. Its rules say where effects go, when they can arrive and who may change the relevant state. A new channel begins at an approved present or future point; it cannot introduce an unnoticed effect into settled history.

A conservative horizon

The **safe horizon** is the next time a field may reach without skipping an interruption. Let c be its current time, T the requested target, h its preferred Beat length, b the next incoming message, q the next scheduled change and w the earliest channel watermark. Then choose the first of those limits:

$$t_{\text{end}} = \min(T, c + h, b, q, w)$$

Omit b or q when absent. The formula selects the next clock time; actual frontier comparisons use (time, microstep) keys. Watermarks are inclusive for those keys, not for unfinished future microsteps at the same clock time. Process due inputs at c before choosing a later endpoint. At a boundary, close the preceding interval, apply inputs in causal order and stabilize immediate consequences before declaring the timestamp final.

Where the limits come from

A closed outgoing-message list can establish w directly. A timetable can prove a bus will not arrive before a known time. Travel speed can prove a pedestrian cannot reach a doorway yet. An instantaneous digital channel may provide no advance warning; the communicating fields then move together or use coordinator-controlled delivery. Any delay must come from an actual world rule.

If cyclic dependencies prevent any field from advancing, the coordinator groups that dependency cycle and resolves it jointly. It does not wait forever for each field to certify the other. Bound zero-time reaction cascades; on exhaustion, expose an unresolved simulation fault rather than advancing the clock past it.

Adaptive step size

Preferred h reflects motion, instability, likely interaction and semantic workload, but the safe horizon always wins. Never impose a minimum step that crosses a due effect. Long intervals can use analytical rules for routine change, with mandatory cuts at expiry, arrival, deadlines and other discontinuities. High-frequency physics remains engine-owned; language-model cadence is chosen around meaningful state and interaction boundaries.

10 How the coordinator accepts a Beat

Only one field may change a particular piece of state at a time. The coordinator therefore grants a temporary **write lease**, meaning permission to update the active entities. The lease includes an ownership version, called an **epoch**. After fields merge or split, a delayed answer carrying an old epoch is rejected. Read-only overlap is safe, but the coordinator still checks that facts used by a decision have not changed meanwhile.

```

ADVANCE_FIELD(F, target)
  PROCESS due inputs at F.cursor in causal microstep order
  contract := ALLOCATE(F, target, available budget)
  safe_end := SAFE_HORIZON(F, contract, boundary watermarks)
  if safe_end cannot advance:
    ADVANCE upstream boundary services or JOIN dependency cycle
    return WAIT or RETRY
  lease := ACQUIRE write authority for required components
  snapshot := READ versioned state at F.cursor
  context := COMPILE(snapshot, contract, agent perspectives)
  candidate := PROPOSE Beat ending at safe_end
  if candidate creates a new boundary earlier than safe_end:
    REGISTER boundary; DISCARD proposal; RETRY with shorter horizon
  RESOLVE actions and world transitions in causal order
  DERIVE perceptions; PROPOSE internal-state changes by perspective
  VALIDATE contracts, causality, read versions and ownership epoch
  if invalid:
    REPAIR and REVALIDATE within retry budget
    if still invalid: RETURN failure without advancing canon
  ATOMICALLY COMMIT Beat + transitions + outbox + new frontier
  UPDATE graph projections and summaries; PUBLISH accepted output

```

10.1 Proposal generation is interruptible

A generation request for 14:23:00-14:24:00 is not a commitment. If a call arrives at 14:23:17, discard or truncate the uncommitted proposal and regenerate to that boundary. Do not interpolate a settled outcome backward from the proposed endpoint. Any accepted prefix must have its own valid preconditions and transition order.

10.2 Save changes and outgoing messages together

The Beat and every message it creates are saved as one accepted operation. A durable outgoing queue retries delivery, while message IDs prevent duplicate application. Graph views, keyword search and semantic-search indexes may update later, but required current facts come from the matching authoritative version. An old semantic-search result can never overwrite current custody or a retracted belief.

10.3 Simultaneous conflicts

An explicit game rule resolves competing actions at the same effective time, for example initiative or a recorded seeded draw. A technical tie-break key can make replay deterministic, but should not accidentally grant gameplay priority to whichever worker completed its model call first. Independent changes need no artificial shared narrative cause.

10.4 Example: two fields meet during a phone call

The player field A has reached an unresolved contact boundary at 14:23:17. Conference field B has committed coarse state through 14:23:00 and prefers 60-second Beats. A coordinator-controlled call becomes due at contact. Conference outputs strictly before contact are sealed; outputs and reactions at contact are settled jointly. No retrospective broadcast before contact remains possible.

Stage	Player field A	Conference field B
Before contact	At 14:23:17; contact not yet resolved	At 14:23:00; optional draft to 14:24
Interrupt	Holds at the contact boundary	Drops uncommitted continuation beyond contact
Catch-up	Does not assume the call is answered	Advances 17 seconds under the sealed boundary contract
Exact contact	Caller initiates call	Phone receives call; perception depends on settings and attention
Joint interaction	Conversation owner writes caller state	Same owner writes minister's conversational state
After call	Returns to local scene cadence	Resumes conference cadence and pending questions

Synchronization aligns time and authoritative dependencies; it does not guarantee that the minister accepts the call. He may be speaking, have muted his phone or instruct an aide to answer. The decision is made from his current accessible evidence, intentions and obligations.

Connect only what the call actually couples

A phone conversation links a few characters and devices, not every street between them. A temporary bridge field can own the shared conversation while unrelated background activity stays in its original fields. If people at the conference react to the call, those participants join the coupled set. A first prototype may merge both whole fields for simplicity, with a clear size limit.

A different case: B is already committed farther ahead

If B is committed through 14:24 while A newly proposes a causal call at 14:23:17, this is not ordinary catch-up. The commit policy has either admitted speculative state as final or accepted a late input. The MVP prevents this at interaction boundaries. For a genuinely late external input, accept it at the current authoritative time or reject it under the latency policy. Rewriting player-observed history is not an invisible synchronization operation.

Uncommitted future proposals can always be discarded. Committed history can only be corrected through an explicit versioned repair policy, not relabeled "provisional" after the fact.

11 Example: three fields meet and separate again

Alice's player field A reaches the open contact boundary at 14:30:12. Bob's autonomous field B is at 14:30:00 with 30-second Beats. Devon's concourse field C is at 14:29:00 with 60-second Beats. All three are approaching Cafe North. A delivery at 14:29:40 changed a doorway's availability, and Devon's effect expires at 14:30:07.

11.1 Find which fields truly interact

Connect two fields when they need to change the same state, perceive one another immediately or make one shared decision. Merely reading the same city summary does not connect them. If A interacts with B and B interacts with C, all three must coordinate even when A and C never meet directly. Distance alone would miss that chain.

11.2 Catch up to a common boundary

Pause affected ownership transfers at 14:30:12. C processes the doorway update at 14:29:40 and the effect expiry at 14:30:07, then reaches contact. B processes its own due inputs. Committed exterior effects constrain catch-up. External channels must be sealed up to the pre-contact frontier; the union resolves contact and its immediate reactions together.

The coordinator acquires leases in a stable order, checks for completed or still-running jobs, increments owner epochs and assigns union U to one owner. A slow model result from B's previous owner is rejected. Shared entities and Beats are deduplicated by identity. The union is a set union of required state, not a concatenation of three prompts.

11.3 Resolve and later separate

U uses two-second Beats with L4 interaction detail. Devon's posture may receive an L5 override because Alice tries to see what he is holding. The surrounding city remains coarse. At 14:30:24, Alice leaves, Bob stays and Devon takes a train. Separation is delayed until direct coupling has ceased for a short configured interval, avoiding boundary thrashing.

```
SPLIT(U, split_time)
  FINISH current Beat and all due messages through split_time
  PARTITION future write scopes using the remaining coupling graph
  ASSIGN one owner and a new epoch to every mutable component
  TRANSFER cursors, active events, commitments, effects and queues
  KEEP shared history by reference; do not clone entities or Beats
  REESTABLISH boundary channels and watermarks between children
  RESCORE each child's detail and cadence
```

The cafe argument remains the same Event-Entity after the split. Alice may retain a belief about Bob while Bob's actual belief changes independently. Physical separation does not end a phone call, obligation or rumor channel; those dependencies survive the handoff.

11.4 Common edge cases

Case	Required behavior
Read-only geometric overlap	Share versioned context; keep independent writers
Two owners try to move one ring	Reject one conflicting proposal; resolve using a declared action rule
A-B and B-C coupling	Coordinate the connected group, not two incompatible pairwise merges
Coarse committed Beat spans contact	Reconstruct permitted intermediate state; preserve committed constraints
Future Beat is only a draft	Cancel it and regenerate to the interruption boundary
Delayed message targets settled past	Apply explicit late-input or repair policy; no silent insertion
Sudden teleport into a dormant area	Hold admission at the boundary until required state is caught up
Union exceeds context budget	Retain invariants; reduce optional detail, partition valid work or slow time
Same timestamp, conflicting actions	Use causal microsteps and declared arbitration, not completion order
Split with in-flight messages	Preserve message IDs, routes, delivery times and deduplication state

11.5 Committed coarse intervals

Suppose a committed minute says the minister denied an accusation but does not specify the exact second. A later request for a camera shot inside that minute can elaborate timing only within constraints. If it introduces a new interrupt that would have changed the denial, it is a counterfactual branch, not refinement. Existing observers, outputs and intermediate commitments restrict what can be filled in.

The stronger design is to settle every externally consequential transition at sufficient temporal precision before exporting it. Internal harmless details may retain broad intervals. If uncertainty over a transition's time can change an interaction outcome, refine it before commitment or preserve a queryable set of possibilities and block decisions that require a single answer.

11.6 No hidden rollback in the MVP

Use conservative commit boundaries for player-facing state. Rollback remains limited to uncommitted candidates. A later system may add explicit speculative branches, but presentation, messages and dependent actions must then participate in its rollback protocol. The current report does not claim that changing one graph edge can undo consequences already shown to several players.

This trades some latency for simpler correctness. A loading transition or delayed world clock is preferable to a ring changing owners retroactively after two players acted on its location.

12 What happens while the player is elsewhere?

Important NPCs can have their own Simulation Fields and continue pursuing goals, relationships and plots off screen. Less important entities can sleep at detailed resolution. Their timers, schedules and outward effects still advance whenever they could influence an active part of the world.

12.1 Devon returns after two weeks

Suppose nobody has visited Devon for fourteen days. The system has not written every conversation, but it has maintained his work schedule, required payments, outgoing messages and other external consequences. Before the player meets him, a catch-up job reads those constraints and creates coarse Beats for the missing time. It adds detail around decisions that matter now.

Gap segment	Catch-up action	Constraint preserved
Days 1-6	Advance routine work and rest by rules	Existing shifts and outward effects
Day 7	Elaborate a previously established missed-shift warning	Employer record and notification time
Day 9	Reconstruct Devon's reaction and a private plan	No new past message to an already settled observer
Day 12	Advance a family conversation at L2/L3	Household availability and known testimony
Current arrival	Promote to exact local state	Travel, custody, effects and open commitments

If the missed shift was never settled but would have affected an employer already simulated through the gap, the generator cannot freely introduce it. It must choose a compatible history, synchronize before that dependency is finalized, or report that the dormant-world contract was violated. Catch-up is constrained generation, not permission to rewrite unseen causes of visible events.

12.2 Locations and unnamed populations

A cafe's opening status follows its calendar and committed exceptions. Reconstruct occupants from schedules, travel and capacity. Do not place a named NPC there while another field owns that NPC elsewhere. Background population may remain aggregate until an interaction promotes an individual to persistent identity. Promotion must preserve aggregate counts and already observed attributes.

12.3 Real-time admission

Start catch-up in the prefetch region using measured latency and likely routes. If it finishes late, simplify unneeded elaboration while retaining required state. If required state is still missing, delay admission or slow simulation. A fast token rate alone is not a real-time guarantee: context processing, retries, validation, dependency closure and concurrent load also consume time.

12.4 Adding detail without rewriting history

Finality and detail are separate. A coarse accepted Beat can settle an outcome while leaving exact wording open. A richly described proposal can still be unaccepted. Anything already shown or perceived becomes a constraint on later elaboration.

The refinement contract

A refinement references a parent Beat and fills declared open detail. It must preserve endpoint state, established intermediate transitions, causal order, outward messages, resource use and every recorded observation. It may narrow an unresolved time interval only when doing so remains compatible with all committed constraints. It may not add a consequential cause that should already have affected another field.

```
Refinement
parent_beat_id, covered_interval, added_detail_records
preserved_transition_ids, preserved_boundary_message_ids
observation_constraints, invariant_checks
application_mode: detail_only | replacement_projection
reconciliation_receipt, new_record_version
```

For a minimal implementation, use detail-only refinement: existing canonical transition IDs remain authoritative, and no parent deltas are reapplied. A more advanced replacement projection may substitute finer transitions, but it must prove that aggregation matches the parent contract and mark exactly one representation as authoritative. Both representations must never update the same inventory twice.

Example of legal and illegal refinement

An L1 Beat establishes that Devon accepted safekeeping of Alice's ring before 14:23:17. A later close-up may add that he glanced at the door, provided no earlier observation contradicts this and the glance changes no settled outcome. It cannot add that he secretly handed the ring to Bob, because custody and later actions already depend on Devon retaining it.

If the player already heard exact dialogue, the dialogue is fixed. If Devon's memory recorded a mistaken phrase, refinement must preserve that subjective record even when the true utterance differs. The objective utterance and the character's memory should not be made identical merely to simplify the retrospective scene.

Reproducibility and repair

Replay means reapplying committed changes, not asking the same model to regenerate them. Store outputs, rule versions and arbitration results. Seeds help explain stochastic choices but do not guarantee deterministic language-model regeneration. Corrections append a superseding record with a reason and dependency impact; they never silently erase what a player was shown.

13 From local scenes to a world-scale plot

A living world needs dramatic organization, but one central author should not dictate every character choice. The proposed hierarchy has local scene storytellers, optional city or faction directors, and a global **Meta-Storyteller**. Important NPC fields create their own developments even when no player is nearby.

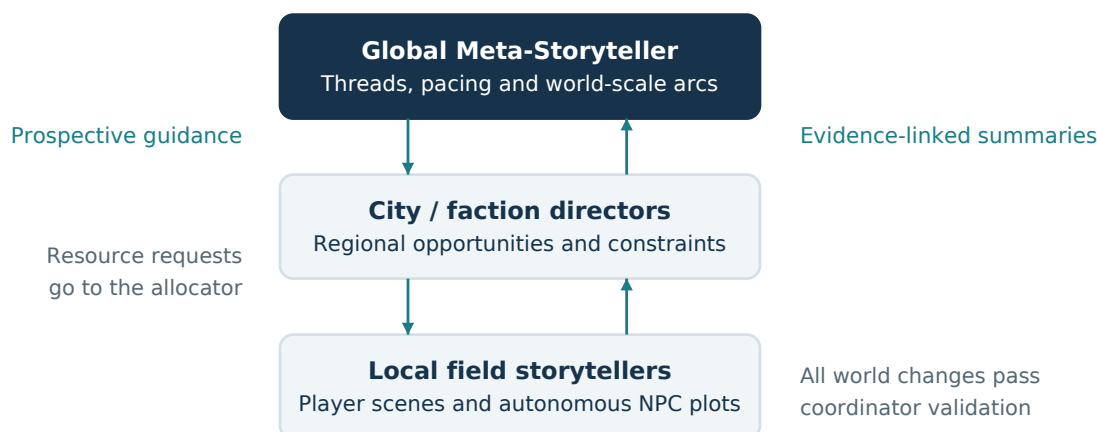


Figure 2. Local fields report what actually happened. Higher layers connect those developments into possible larger plots and send suggestions downward. Every suggestion still passes ordinary world validation, and private character knowledge remains private.

13.1 What local fields report upward

Local fields publish short, evidence-linked summaries: the time covered, source Beats, changed promises and relationships, open decisions, outward consequences and possible dramatic importance. Higher levels remove duplicates and record which areas are stale. A city summary must not pretend every district is current when some have not advanced.

Safety-critical constraints and scheduled effects use structured records, not lossy summaries. A high-level report may mention political tension; the exact time a curfew becomes effective must remain independently available.

13.2 What higher storytellers may propose

Directors track open questions, promises, possible payoffs, pacing and opportunities for separate plots to meet. They may request more compute for a dormant NPC, suggest a future meeting or ask for more detail about a negotiation. The Attention Allocator still decides what fits the budget, and the coordinator still protects time and accepted history.

A director may influence environmental opportunities within a granted authorial scope. It cannot directly overwrite an agent's beliefs, guarantee that a player accepts a quest or make an already committed failure disappear. An NPC's refusal can change the plot. Narrative coherence is assessed after genuine consequences, not enforced by making every intention serve the preferred ending.

13.3 Direction contracts and a political example

Directive type	Meaning	Authority limit
World constraint	An authored rule or established restriction	Cannot contradict stronger canon or game rules
Scheduled proposal	A proposed future occurrence with prerequisites	Becomes authoritative only through validation and commit
Dramatic pressure	A desired development such as rising tension	Does not specify mandatory character behavior
Opportunity	A meeting, clue route or complication may be offered	May be ignored, refused or prevented
Forecast	An expected trajectory or possible future	Never treated as a fact that already happened
Protected decision	Choice reserved for a player or author	Local and meta-storytellers may not decide it

At 14:35, the city director receives summaries of the conference denial, Devon's warning and Alice's confrontation. It identifies a potential connection: Devon's employer holds a municipal contract relevant to the accusation. That connection is usable only if established in canon or validly introduced within an open authorial scope. Semantic convenience is not evidence.

The director proposes an opportunity for a journalist to visit the employer after 15:00, with prerequisites, expiry, involved entities and a compute ceiling. The journalist's agent then chooses whether to act from its goals and accessible information. If it proceeds, travel, interview and information transmission create ordinary Event-Entities and Beats. Devon may refuse to talk. Alice cannot learn the result until a valid communication reaches her.

A typed directive

```
Directive
  id, issuer_scope, target_field_or_thread, type
  issued_at, earliest_effective_time, expires_at
  prerequisites, desired_development, allowed_interventions
  protected_choices, prohibited_changes, resource_ceiling
  source_beat_refs, approval_status
```

Hierarchy without competing world copies

Faction directors and city directors may cover the same agents. Their scopes are scopes of guidance, not separate ownership of those minds. Contradictory requests are resolved by authority, author settings and resource priority; unmet guidance is reported. A faction director's authorial knowledge is distinct from that faction's in-world intelligence records.

Macro models can advance weather, resources, transport or economic constraints. The Meta-Storyteller interprets those developments and shapes opportunities around them. Neither a weather model nor an authorial pacing target may independently change a locally owned state component without a coordinated transition.

14 Data contracts for a prototype

These logical records provide a starting schema. Stable IDs, temporal scope and provenance matter more than database choice; psychological categories need not each become a table.

Record	Minimum fields and relations
Entity	ID, kind, aliases, lifecycle, component schema, current owner
EventEntity	ID, type, participants and roles, phase, start/end bounds, parent events, active commitments
Beat / Change	Section 4 contract; links to all affected entities and event IDs
FactVersion	Subject, predicate, value/reference, perspective scope, valid interval, record version, source Change
Perception	Observer, modality, content, world interval, attention/reliability, accessible source refs
MemoryTrace	Holder, content, source percepts, encoding time, salience, retention/revision state
Recollection	Holder, retrieval time, reconstructed content, trace refs, distortion/confidence annotation
BeliefVersion	Holder, proposition ID, stance, confidence band, valid interval, evidence and contradiction refs
Proposition	Atom or mental-operator expression, referenced entities, attributed holder, nesting depth
BoundaryMessage	ID, source/target, emission/delivery time, causal parent, payload, delivery and deduplication status
Field	Anchors, active/read/dependency sets, ownership epoch, cursor, detail frontier, policies, pending queues
Summary / Thread	Scope, coverage interval, source IDs, freshness, open questions or narrative commitments

14.1 Provenance is a graph too

Beliefs reference accessible evidence, memories reference perceptions, and perceptions reference accepted stimuli. Seeded knowledge is marked initialization, not justified with fabricated experience. Inferences can reference prior beliefs, but circular sources cannot substitute for evidence.

14.2 Query contracts

Required queries return entity state at t , changes over an interval, beliefs and supporting evidence, component owners, and incoming-channel watermarks. Answers include version, precision and coverage. "Not modeled" is not proof that nothing happened.

Participation and epistemic graphs are separate views with shared identity, not independently edited biographies.

14.3 Example records with explicit perspective

```

EventEntity conference_42
  kind: press_conference
  participants: minister, spokesperson, journalists
  phase: questions
  interval: [14:00, expected 15:00]
  open_issue: response_to_payment_accusation

Perception percept_A17
  observer: Alice
  time: 14:22:05
  modality: hearing
  content: fragment "authorize the payment"
  missing_audio: true
  source: utterance_U8

BeliefVersion belief_A9
  holder: Alice
  proposition: authorized(minister, payment)
  stance: suspected
  confidence: medium
  evidence: percept_A17, prior_belief_A2
  valid_from: 14:22:06

BeliefVersion belief_A10
  holder: Alice
  proposition: BELIEVES(Bob, authorized(minister, payment))
  total_depth_including_holder: 2
  evidence: Bob_statement_U9
  note: does not assert Bob actually believes this

```

The simulation may establish Bob's actual belief separately and find it disagrees with Alice's attribution. Likewise, a later correction to belief_A9 closes its validity interval and opens another version. It does not delete percept_A17 or retroactively improve the audio.

Effects and component ownership

```

Effect effect_D4
  target: Devon
  type: fictional_sedative
  onset: 14:00:07
  expires_at: 14:30:07
  modifiers: attention_profile = distracted
  stacking_rule: replace_same_type
  source: accepted Change D4
  expiry_owner: world effect service

```

A local storyteller can read this effect and use its perception consequences. It cannot extend the duration merely to make a scene interesting. The authoritative effect service schedules expiry and notifies the component owner, including across field merges and splits.

15 Minimal viable prototype

Build a small text-first city with three locations, six named agents, a few persistent objects and three Event-Entities: a cafe exchange, a press conference and Devon's work conflict. Add two simulated player inputs to exercise simultaneous interaction. Use one coordinator, one authoritative store and a durable boundary queue. A graph database and multiple parallel model workers are optional.

15.1 Scope the first implementation

Support all six level labels as contracts, but test L1, L3 and L4 first. Higher-order beliefs can reach depth three only in explicit scenarios. Use a simple rule-based spatial resolver, typed relation expansion and a fixed scoring policy. The local generator returns structured Beat proposals; code validates IDs, intervals, possession, access and message timing. Psychological plausibility is reviewed separately from structural validity.

```
COORDINATOR_CYCLE(target_time)
  INGEST player actions under the authoritative input-time policy
  UPDATE boundary service and deterministic timers
  RESOLVE active fields and expected near-term contacts
  RANK due fields; reserve budget for interrupts
  for each causal interaction group:
    SEAL source outputs to the next safe boundary
    MERGE coupled write scopes when necessary
    ADVANCE_FIELD(group, target_time)
    SPLIT only after coupling ends and handoff is complete
  UPDATE regional summaries from committed Beat IDs
  RUN directors if their cadence or trigger is due
  VALIDATE and schedule prospective directives
```

15.2 Practical simplifications

Use whole-entity ownership at first; component ownership can come later. Model only registered channels: physical movement, nearby speech, calls and one news feed. Before an active group advances, run the connected sources' coarse boundary passes. With zero lookahead, group them under the same coordinator. No distributed speculative rollback is required.

Keep canonical records transactional; implement graph views with indexes on entity, relation, valid time and Beat ID. If a retrieval index lags, fall back to exact state and source records. Persist accepted model outputs and versioned prompts for reproducibility. On a model timeout, do not advance social canon with an empty successful Beat; retry within budget, apply an authorized deterministic fallback or hold the boundary.

The prototype succeeds when its transitions and synchronization are inspectable, not when it generates the most prose. Start by asking it to explain every consequential current state through a short chain of accepted Beats and evidence.

15.3 Implementation milestones and acceptance tests

Milestone	Deliverable	Acceptance condition
1 Canonical ledger	Entities, Beats, transitions and snapshots	Replaying accepted changes reproduces current state
2 Perspective model	Perceptions, memory and beliefs	Alice cannot use an unheard private statement
3 Adaptive fields	Typed resolver and semantic contracts	Remote call adds the minister without loading the whole city
4 Causal synchronization	Boundary queue and sealed horizons	No committed reaction precedes message delivery
5 Collision handling	Union ownership and split handoffs	Three-field merge preserves unique custody and rejects stale owners
6 Catch-up	Coarse progression plus detail refinement	Dormant history respects every prior outward effect
7 Narrative hierarchy	Local summaries and city/global directives	Guidance creates opportunities without rewriting beliefs or player choices

Required deterministic regression scenarios

Run the same scripted inputs with two-second player Beats and 60-second remote Beats. Insert a call at second 17, a scheduled effect expiry at second 7 and an object handoff contested by two agents. Assert exact delivery order, one authoritative custodian and complete effect processing.

Next introduce A-B-C coupling with three different cursors, merge at a common boundary, then split while a message remains in flight. Assert one owner per mutable entity, no duplicate message application, preserved Event-Entity IDs and continuous history. Submit a delayed result with an obsolete owner epoch and confirm rejection.

Finally, request a detail-only refinement of a committed coarse Beat. Its prose may expand, but inventory, boundary messages and established perceptions must not change. Attempt an illegal refinement that adds a prior secret transfer and confirm rejection. A model's fluent explanation is not an acceptance criterion.

Agent-level regression scenarios

Test partial hearing, misleading testimony, correlated rumor sources, forgetting, later correction and mistaken second-order belief. Verify each revised belief cites only agent-accessible evidence. The system must be able to answer both "Where is the ring?" and "Where does Alice think Bob believes it is?" without conflating the scopes.

For autonomous plots, leave the minister outside every player field and verify that scheduled Beats still advance his event and goals. Then connect a player and verify the result uses that established history rather than inventing a replacement backstory at contact.

16 Evaluation and resource behavior

No empirical results are claimed. The evaluation program should separate hard invariants, approximation quality, subjective narrative quality and operating cost. A beautiful scene that violates ownership is a failure; a structurally valid scene can still be dull or psychologically implausible.

Dimension	Measure
Canonical integrity	Temporal, ownership and duplicate-application violations
Epistemic integrity	Belief updates without accessible evidence; perspective leakage
Synchronization	Late causal effects, contact alignment, rejected stale jobs, blocked time
Approximation	Outcome differences between coarse and fine reference simulations
Retrieval	Required-component coverage, relevant-history recall, stale-index use
Resource allocation	Tokens, calls, wall latency, queue wait and memory per field
Narrative quality	Human ratings of agency, coherence, pacing and meaningful surprise
Autonomy	Progress of important NPC goals while players are elsewhere

16.1 Compare policies rather than prose volume

Useful ablations remove entity-centered state, perspective isolation, typed expansion, attention allocation, sealed boundary horizons or hierarchical direction one at a time. Compare fixed detail against adaptive detail under equal total compute. Test whether adding the Meta-Storyteller improves long-range coherence without reducing player or NPC agency.

Coarse and fine runs need not produce identical fiction, but should preserve rules and comparable outcome distributions. Use controlled seeds for mechanics and store generated proposals; text regeneration is not the replay target.

16.2 Resolution bias and promotion consistency

More Beats must not give important characters more chances to act. Normalize hazards, needs and action opportunities by world time, not call count. Otherwise sixty one-second updates can create far more accidents or rumors than one minute-long Beat.

Promotion should preserve identity, active commitments and aggregate constraints. Demotion should not erase inconvenient beliefs. Monitor oscillation around field boundaries, repeated history invention, and systematic differences between NPCs with high and low compute budgets. The allocator's policy can change the simulated society, not only its cost.

16.3 No throughput guarantee from token rate

Measure end-to-end contact latency, including context construction, model prefill, generation, validation, retries and dependency synchronization. Admission policies must be based on measured bounds or explicit failure handling. Report resource savings alongside how often the system reduced detail, deferred interaction or slowed world time.

17 Limits and outlook

The hardest remaining problem is deciding which omitted details can later become consequential. The architecture handles this with declared component contracts, sealed external effects and constrained refinement. Those mechanisms make the risk visible; they do not provide an oracle for narrative relevance. An unexpected player action can expose a detail that was never modeled.

17.1 Boundaries of the model

Mechanical validators can verify object identity, time intervals and message order. They cannot certify that an emotional transition is psychologically true. A coherent false-belief trace demonstrates consistency with a fictional agent model, not human Theory of Mind. All cognition profiles should remain inspectable and configurable rather than presented as universal laws.

Character profiling in real environments is a different application. Observed statements, inferred beliefs and sensitive mental-state hypotheses require consent, purpose limits, correction and retention controls. Nothing in a game-oriented belief model establishes reliable access to a real person's thoughts.

17.2 Interactive film and multiplayer

For interactive film, committed Beats can feed camera, dialogue, sound and scene-generation systems. A narrator can show an omniscient view while individual characters retain limited knowledge. For multiplayer, one shared event can produce different perception records for several players and NPCs. Client presentation may predict motion, but authoritative narrative consequences require reconciliation through the coordinator.

At larger scale, regions may become separate services. Sharding must preserve global identity, boundary delivery and ownership epochs. A director hierarchy can scale summaries, but it cannot replace causal synchronization with approximate prose. Begin with a world small enough that every cross-field effect is inspectable, then distribute the same contracts.

17.3 Conclusion

The Living World Model is a proposal for persistent consequences under selective computation. Event-Entities provide continuity for scenes and processes. Beats record their ordered effects across physical, social and mental state. Entity-centered views expose personal and object histories without fragmenting canon. Perspective-specific memory and belief preserve the difference between what happened, what was experienced and what is now believed.

Adaptive fields decide where detail is needed; boundary contracts decide when it is safe to advance. Important NPCs maintain their own lives outside player range. Meta-storytellers connect those lives into larger dramatic possibilities while leaving outcomes subject to world rules and agent choices. These separations give a prototype a concrete objective: a world whose past can be explained, whose present is shared, and whose future is still genuinely open.

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Status and terminology

Living World Model, Beat profiles and the proposed allocation and synchronization contracts are the terminology of this report. No claim of priority over all related work is made. The six semantic levels, scoring weights, radii, times, budgets and prototype scenarios are illustrative design choices to be tested. References motivate adjacent concepts; they do not empirically validate the full architecture.

Prototype handoff summary

Build the authoritative Beat ledger and deterministic tests first. Add separate agent perspectives, then variable-duration fields with conservative boundary handling. Exercise a remote call and a three-field merge/split before attempting a large world. Add narrative direction only after local state and causal history survive those tests. The smallest convincing demonstration is not a giant city. It is a small city in which every important consequence remains consistent when the player leaves, returns or interrupts somebody else's story.